

TUSHAR GARG

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Software Engineer | React • FastAPI • PostgreSQL • AWS • TypeScript

Full Stack and Backend Engineer with hands-on experience building and deploying production-grade web applications, REST APIs, and cloud-native backend services. Experienced end-to-end — from system design and API development through containerised deployment on AWS with Docker and CI/CD. Skilled in FastAPI, React, PostgreSQL, and TypeScript, with additional experience integrating ML-powered features into full-stack applications.

SKILLS

Frontend: React.js, TypeScript, JavaScript, HTML5, CSS3

Backend & Systems: FastAPI, Spring Boot, REST APIs, PostgreSQL, MongoDB, API Design, System Design, Node.js, Express.js

Cloud & Infrastructure: AWS (EC2, RDS, S3), Docker, GitHub Actions, CI/CD, Nginx, Linux

Programming: Python (primary), Java, C/C++, JavaScript, C#, PHP

AI & ML: Machine Learning, Computer Vision (CNNs), Explainable AI (SHAP), Feature Engineering, Scikit-learn, NumPy, Pandas

Languages: English, Hindi, Japanese (JLPT N3)

EXPERIENCE

AI / Manufacturing Systems Intern | ZenKensa Project

Dec 2025 – Feb 2026

- Developed computer vision-based defect detection workflows using Python, OpenCV, and ML models, enabling automated surface inspection for industrial manufacturing lines.
- Built inspection report automation and AI-assisted decision-support tooling in Python, reducing manual inspection effort ~20%.
- Contributed to deployment packaging and on-site usability testing, ensuring the system operated reliably in an industrial environment outside a controlled lab setup.

IT & Data Analysis Intern | Anant Raj Cloud Pvt. Ltd.

May 2024 – Jun 2025

- Implemented environmental monitoring and risk analysis systems using Python, FastAPI, and ML-based analytics pipelines across healthcare, industrial, and residential domains.
- Developed modular backend services and reporting pipelines supporting multi-parameter environmental monitoring, automated risk scoring, and structured decision-support outputs.
- Contributed to risk analysis reporting infrastructure, translating domain requirements into well-defined backend services with reliable, structured data outputs.

PROJECTS

FinKizuna ERP Platform — FastAPI, PostgreSQL, AWS, Docker, CI/CD, Isolation Forest, SHAP [\[GitHub\]](#)

- Designed and deployed a multi-tenant ERP platform on AWS (EC2, RDS, S3) using FastAPI and PostgreSQL, supporting user authentication, workflow automation, and structured reporting.
- Implemented ML-based fraud detection engine (~94% accuracy) using Isolation Forest with SHAP-based explainability, producing transparent, auditable risk scores per transaction.
- Built end-to-end CI/CD deployment pipelines using GitHub Actions, Docker, and Nginx — enabling automated, repeatable deployments to production with environment-based configuration.

Explainable HIV Drug Resistance System — Python, Machine Learning, SHAP, FastAPI, Bioinformatics [\[GitHub\]](#)

- Developed an explainable ML system for predicting HIV drug resistance from genomic sequence data, integrating SHAP for deterministic, interpretable model outputs.
- Integrated ML inference with FastAPI to serve predictions, producing per-feature SHAP attributions for each resistance classification to support interpretable outputs.

VERITAS — AI Indoor Environmental Risk Intelligence System — Python, Machine Learning, FastAPI, Data Pipelines

- Built an AI-driven risk scoring system that ingested and analysed multi-parameter environmental sensor data, producing automated risk scores and structured reporting outputs.
- Designed modular data pipelines using Pandas/NumPy covering ingestion, multi-parameter analysis, automated risk scoring, and structured report generation — developed independently end-to-end.

EDUCATION

B.Tech. in Biotechnology — Amity University, Noida

2022 – 2026 | CGPA: 8.6

CERTIFICATIONS

AI & Machine Learning Core | Python for Data Science | Full Stack Web Development | Version Control with Git & GitHub |

Microsoft Technical Certification | Data Analysis & Excel (Udemy)